

Lecture week 4: Reinforcement Learning

Iris Proff

Introduction

Video 1: Reinforcement Learning

Video 2: Modelling behavior with RL

Ways to model cognition

Structural model

qualitative description
of cognitive or perceptual processes

Cognitive model

computational or mathematical description
of cognitive processes

Neural model

computational or mathematical description
of neural processes

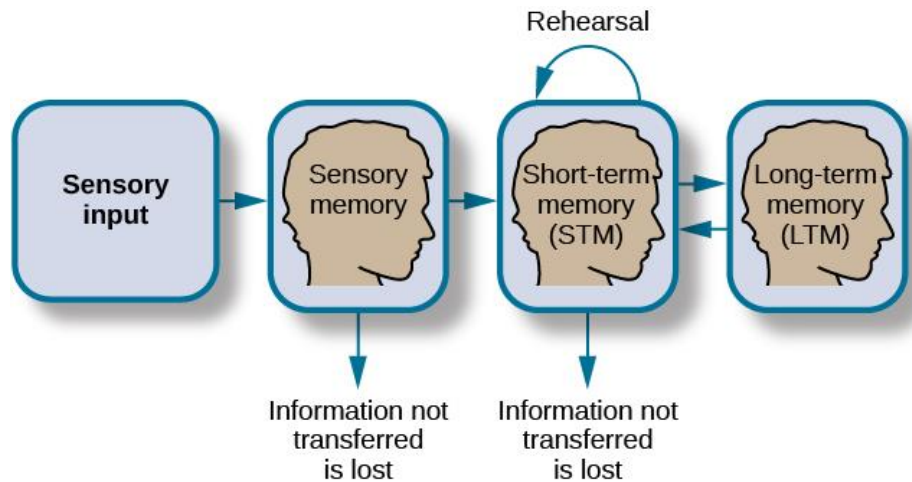
Abstraction



Three memory models

Structural model

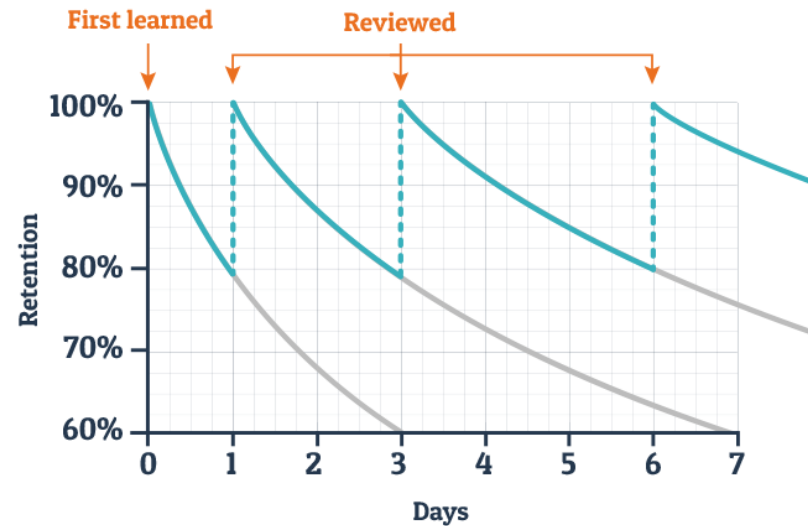
describes how different kinds of memory interact



Atkinson and Shiffrin (1968)

Cognitive model

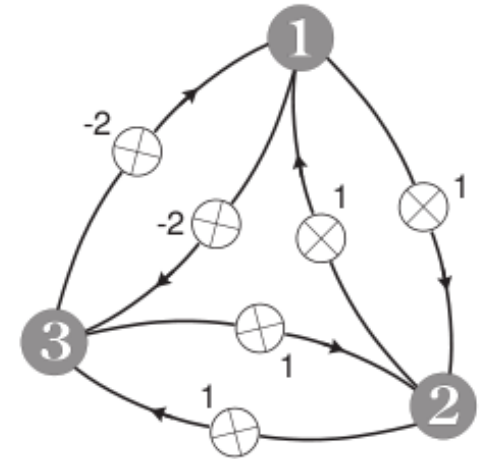
models temporal course of memory decay



Ebbinghaus (1885)

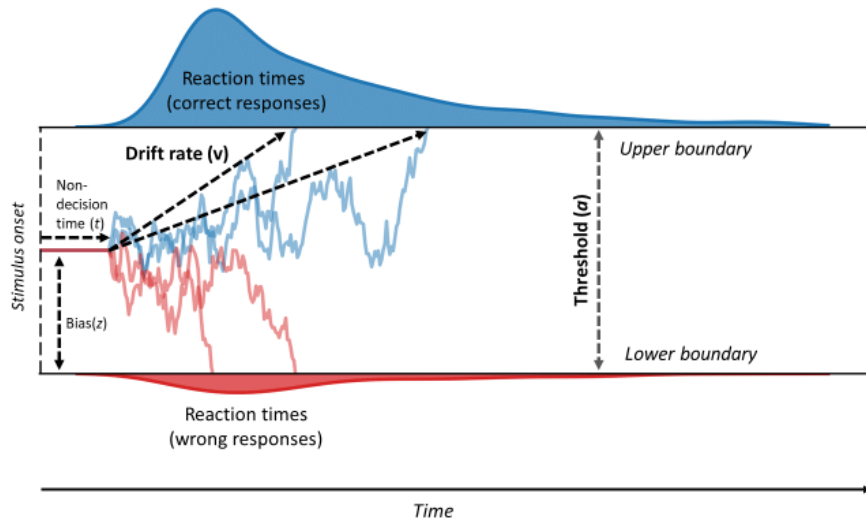
Neural model

models how a neural network stores and retrieves one pattern



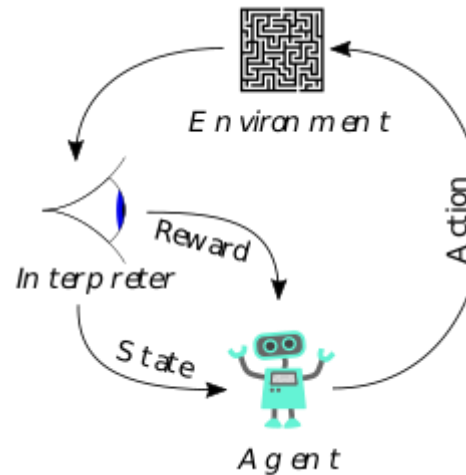
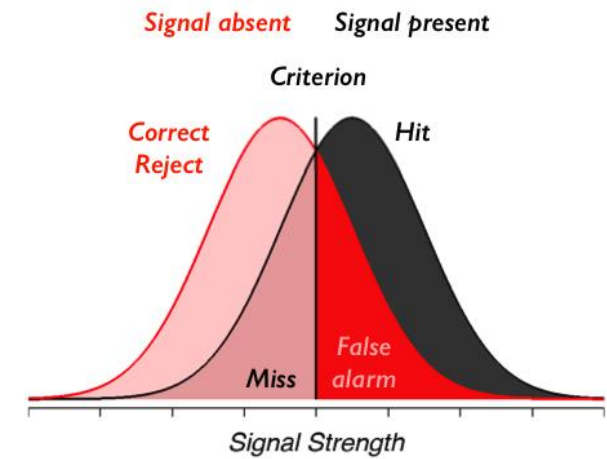
Hopfield (1982)

Some cognitive models



Drift diffusion model

Signal detection theory



Reinforcement learning

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Video 1: Reinforcement Learning

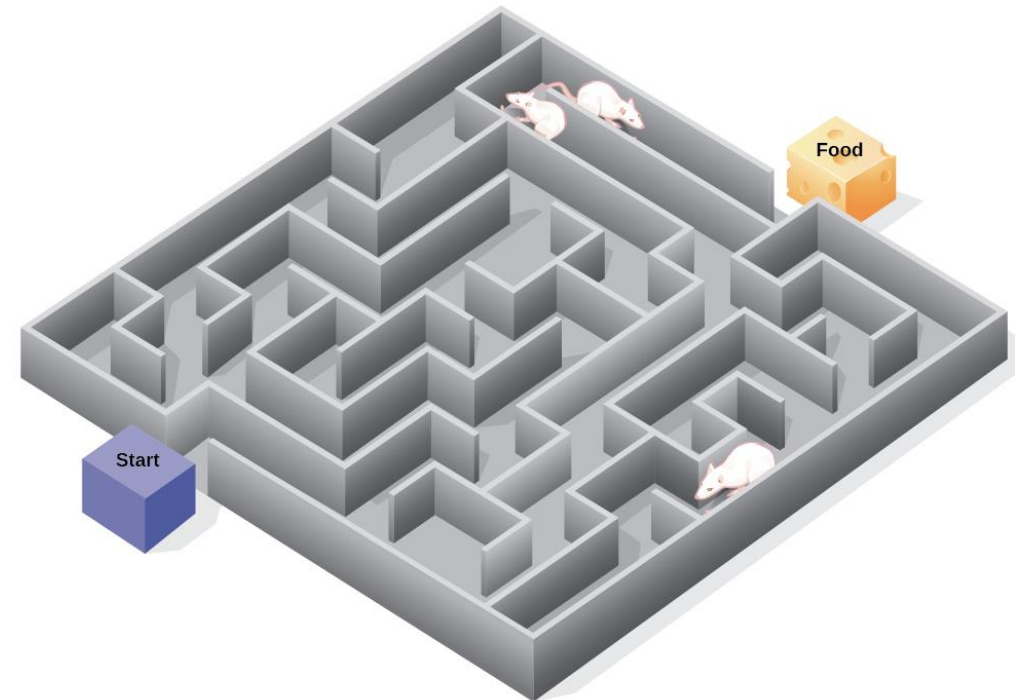
Video 2: Modelling behavior with RL

Reinforcement learning

How does an agent learn which action is best in a given situation...

- in an **uncertain** environment
- without a **teacher** signal
- but in presence of (delayed) **feedback**?

RL means learning by interacting with the environment



Types of learning algorithms

- Supervised learning

Learning from labeled examples – neural networks

- Unsupervised learning

Finding patterns in the raw data - clustering

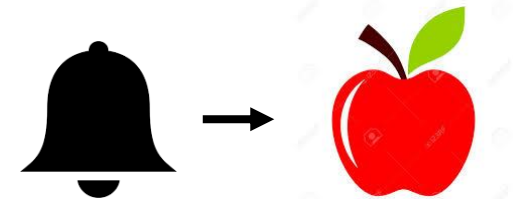
- Semi-supervised learning

Learning from reward signals – Reinforcement learning

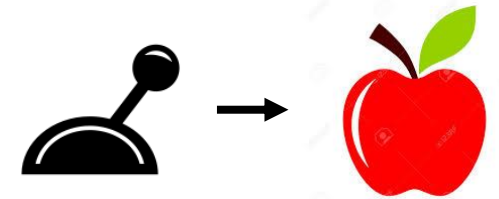
Reinforcement learning

A structural model from behavioral psychology...

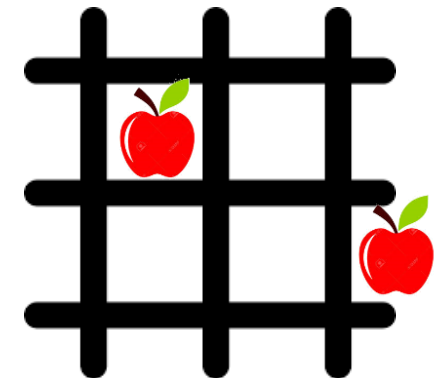
Pavlov: 1897 Pavlovian learning / classical conditioning
→ agent learns stimulus-outcome contingency



Thorndike: 1905 Law of effect / instrumental conditioning
→ agent learns action-outcome contingency



Tolman: 1930 Latent learning / goal directed learning
→ agent learns cognitive map



Reinforcement learning

...formalized by AI to a cognitive model

Structural model

Classical conditioning

Instrumental conditioning

Goal directed learning

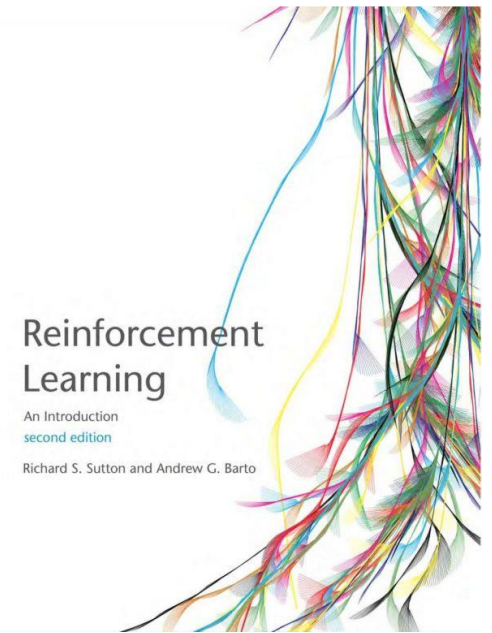
Cognitive model

→ Rescorla-Wagner model (1972)

→ Temporal difference learning (Samuel 1959)

→ Q-Learning (Watkins & Dayan 1992)

→ Model-based learning



Determining utility (optimal control theory)



$$U(S) = R(S)$$

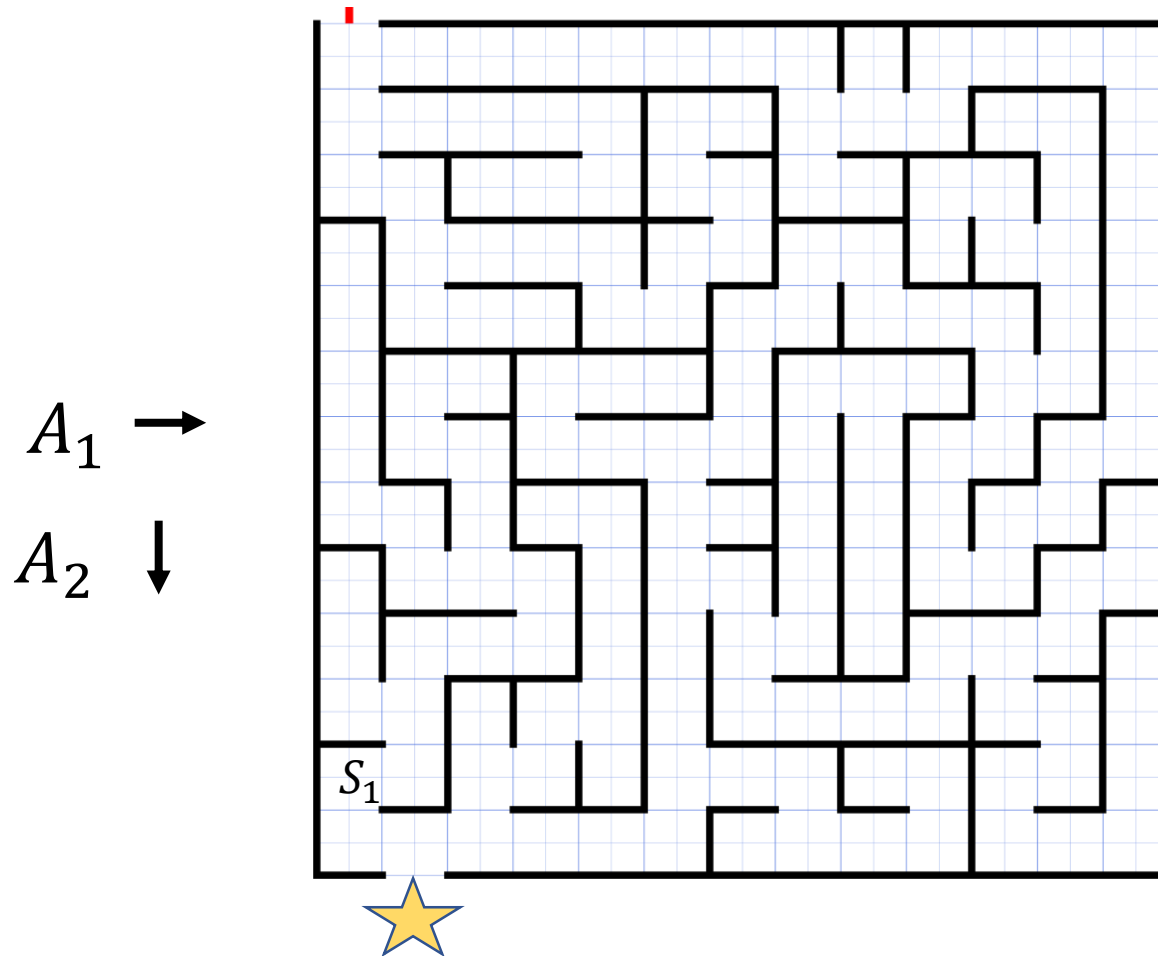


R_1



R_2

Determining utility (optimal control theory)



Bellman equation:

$$U(S) = R(S) + \gamma * \max_A [U(S')]$$

U: utility

S: state

R: reward

A: action

γ : temporal discount factor

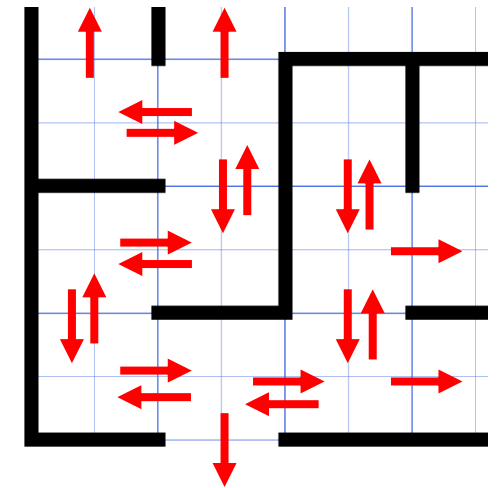
Q-Learning / temporal-difference learning

How to learn the utility of state-action pairs by trial-and-error?

Learning the Q-value = approximation of utility

$$Q(S, A) = \hat{U}(S, A)$$

Online learning (policy is applied and learned at the same time)



Q-Learning – update rule

Prediction error

$$\delta = R + \gamma * \max_{A'}(Q_t(S', A')) - Q_t(S, A)$$

S: state

R: reward

A: action

γ : temporal discount factor

α : learning rate

U: utility

Q-value update

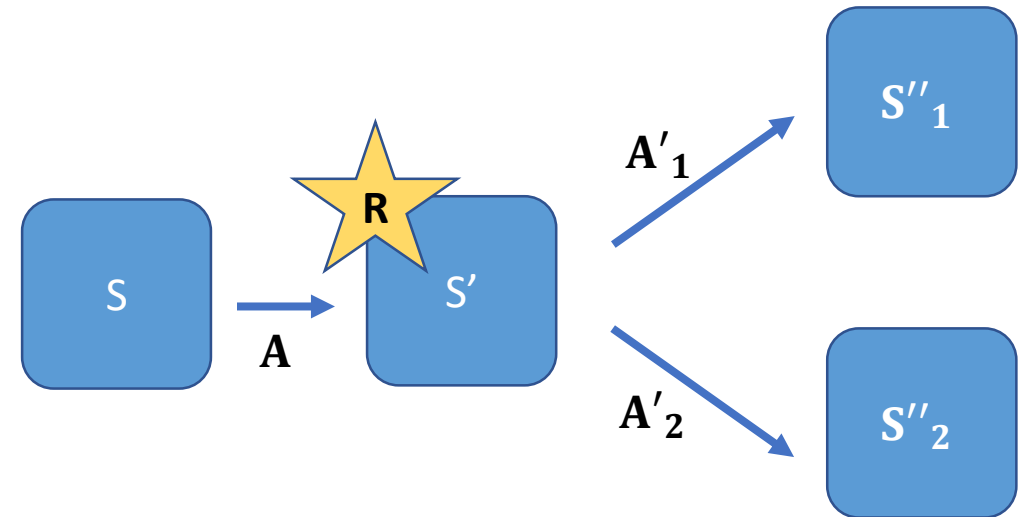
$$Q_{t+1}(S, A) = Q_t(S, A) + \alpha * \delta$$

Q-value approximates utility

$$\lim_{t \rightarrow \infty} Q_t(S, A) = U(S, A)$$

Reminder: Bellman equation

$$U(S, A) = R(S, A) + \gamma * \max_{A'}[U(S', A')]$$



Q-learning – action selection policy

Say we are in state S , with possible actions A_1 and A_2 , with
 $Q(S, A_1) = 0.5$ and $Q(S, A_2) = 0.7$

Which action should we choose?

Exploration – exploitation trade-off



exploit what you know

should you...

or

explore new options?



“A reinforcement learning agent has to exploit what it has already experienced in order to obtain reward, but it also has to explore in order to make better action selections in the future.

The dilemma is that neither exploration nor exploitation can be pursued exclusively without failing at the task.”

Q-learning – action selection policy

Say we are in state S , with possible actions A_1 and A_2 , with
 $Q(S, A_1) = 0.5$ and $Q(S, A_2) = 0.7$

Which action should we choose, to trade off exploration and exploitation?

Epsilon greedy:
$$P(a_i|S) = \begin{cases} \frac{1-\epsilon}{|max_{qi}(A)|} & \text{for } q_i = max(A) \\ \frac{\epsilon}{|A|-|max_{qi}(A)|} & \text{else} \end{cases}$$
 best action chosen with $P = 1 - \epsilon$

Softmax:
$$P(a_i|S) = \frac{e^{\frac{q_i}{\tau}}}{\sum_A e^{\frac{q_i}{\tau}}}$$
 for $\tau \rightarrow 0$ greedy, for $\tau \rightarrow \infty$ random

Q-Learning – pseudo code

Set $\alpha \in (0,1]$, small $\varepsilon > 0$, $\gamma > 0$

Initialize $Q(s, a)$ for all $s \in S, a \in A$ to 0

Initialize s

For number of episodes

 While s is not a terminal state

 choose a from s using action selection policy

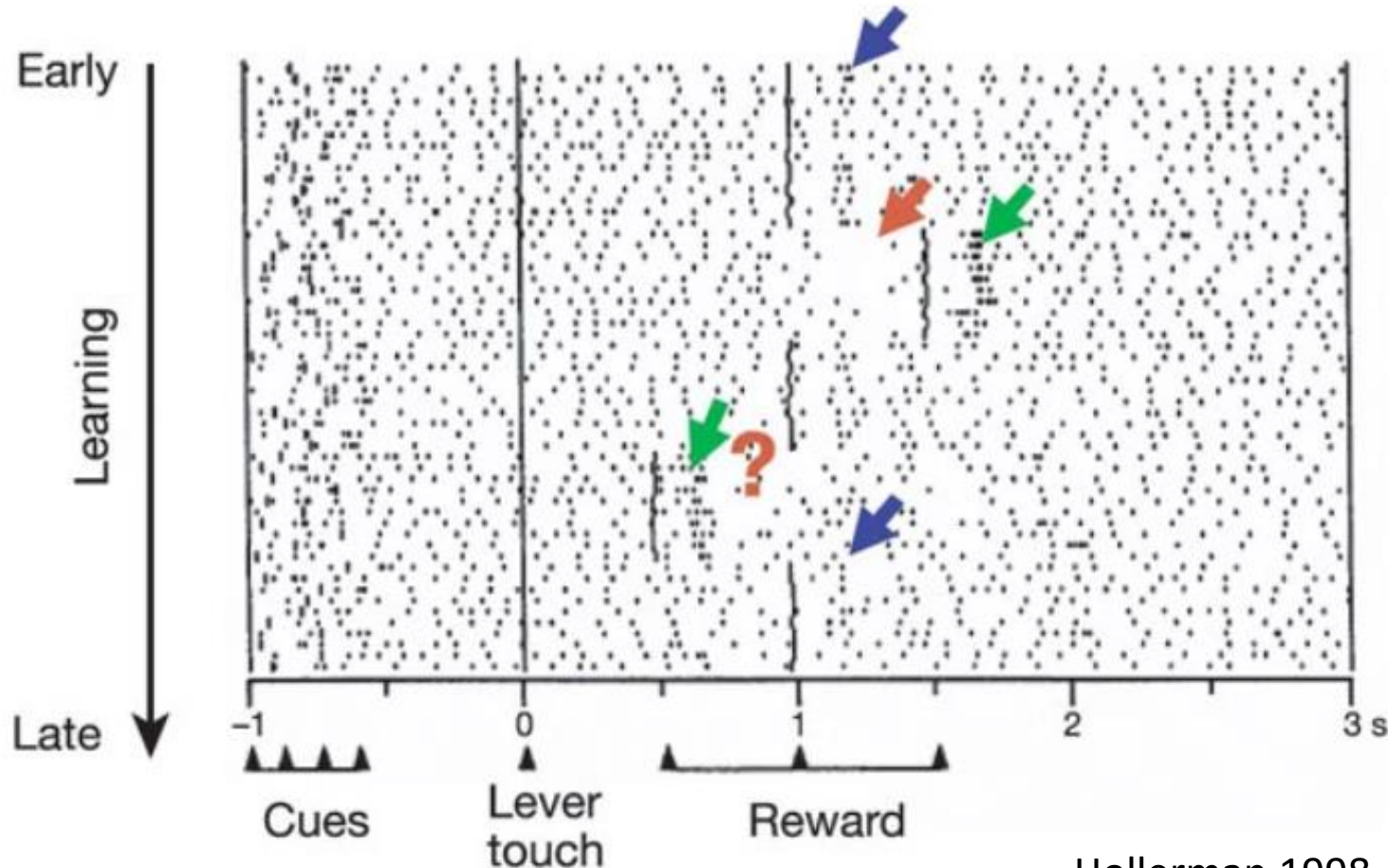
 observe R and s'

 update Q

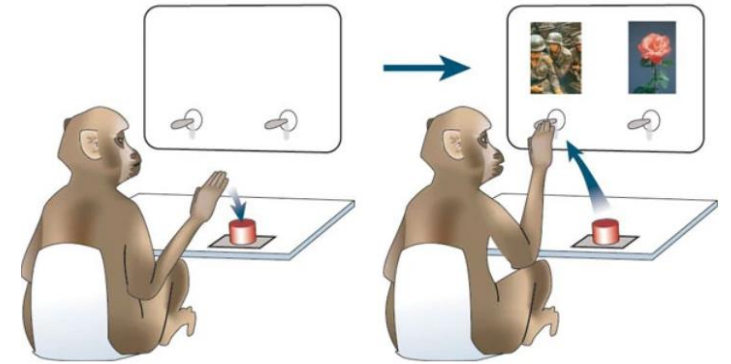
 update s to s'

Temporal difference learning in the brain

Recording from a single dopaminergic neuron in monkey brain

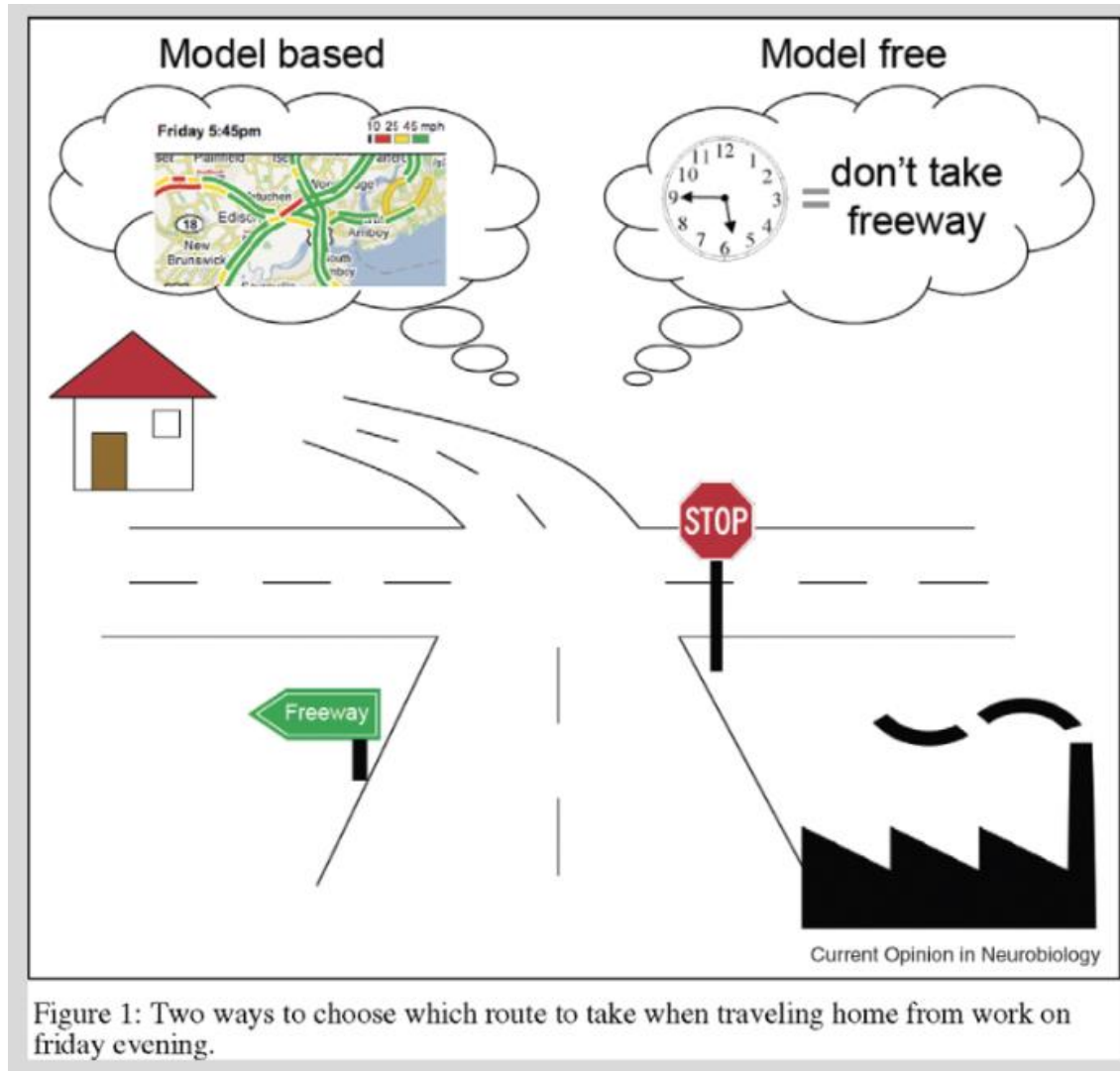


Hollerman 1998



Upper blue: expected reward
Red: expected reward omitted
Green: unexpected early/late reward
Lower blue: expected reward

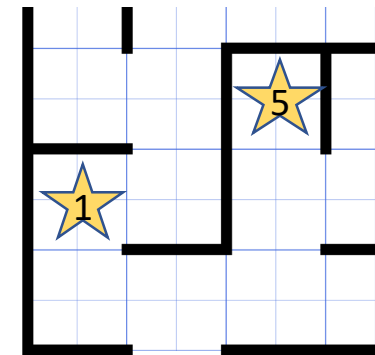
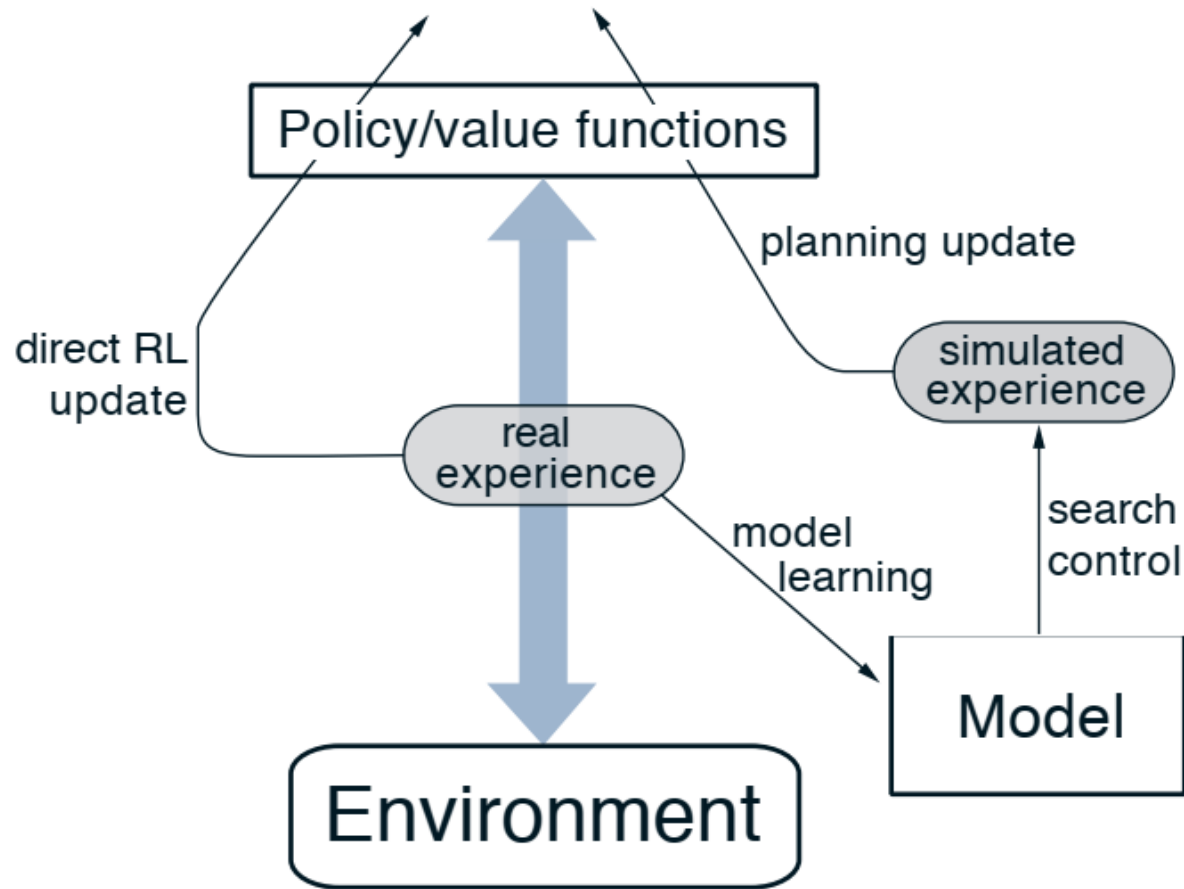
Model-based vs model-free learning



Model-free (habitual): rely on trial-and-error, efficient but inflexible

Model-based (goal directed): build a model of the environment, effortful but flexible

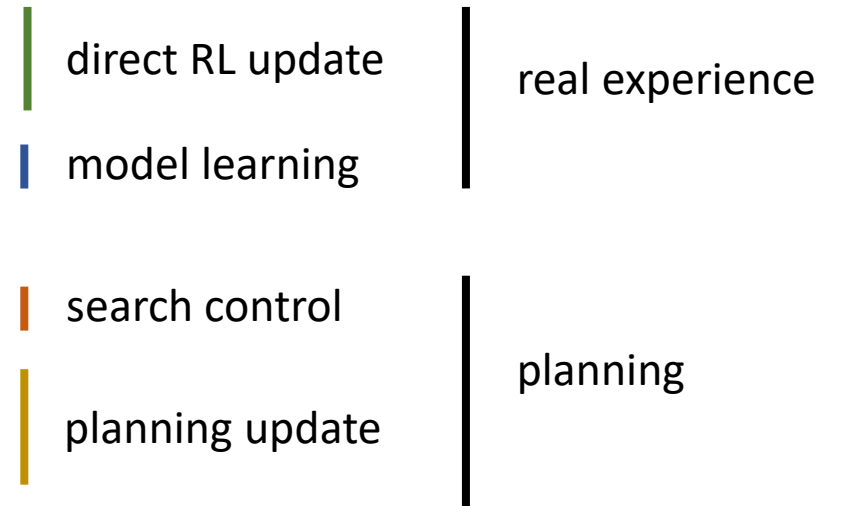
Model-based reinforcement learning



Model: $(s, a) \rightarrow (s', r)$
»What does my environment look like? «

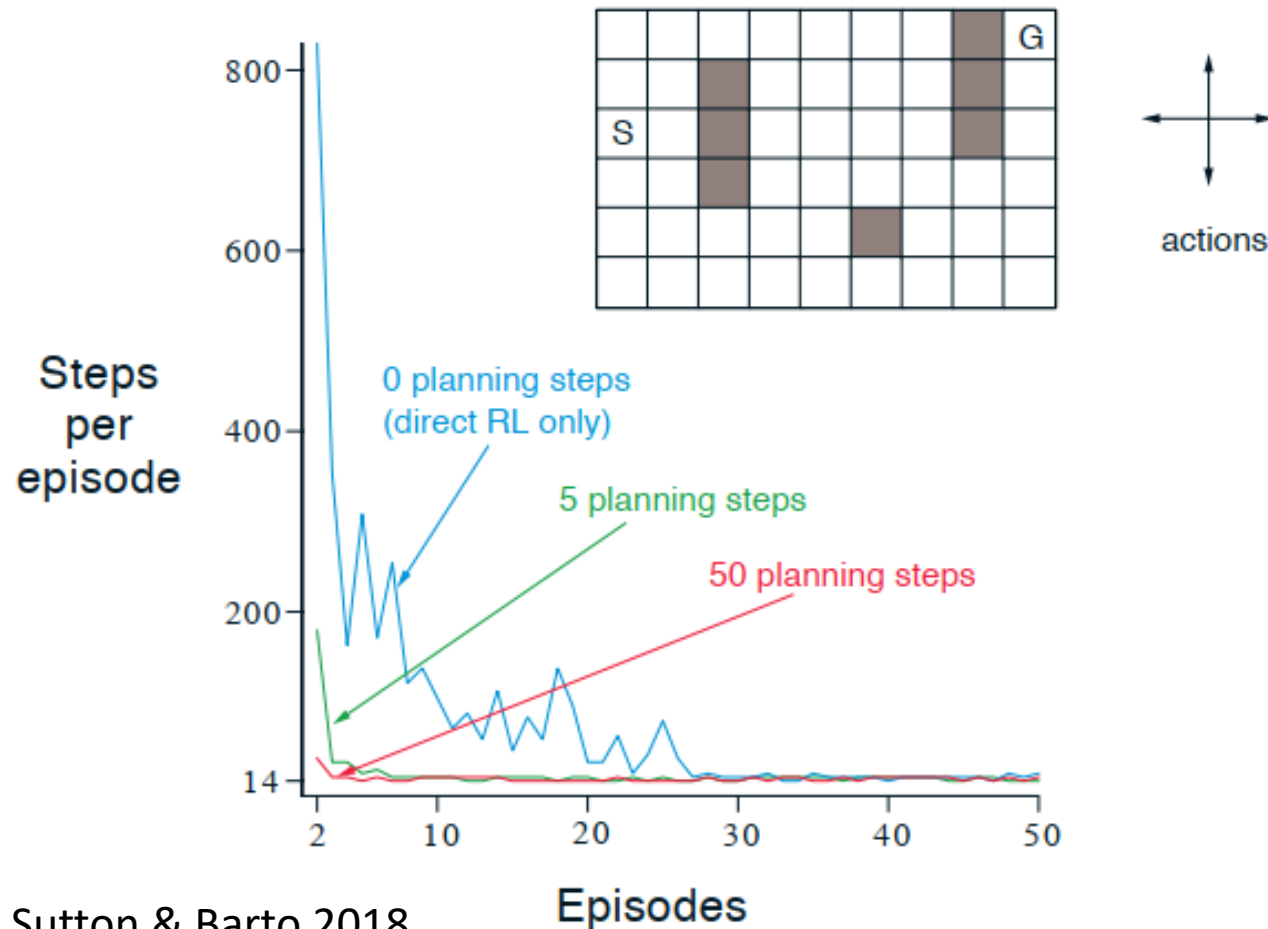
Dyna-Q – pseudo code

```
initialize parameters
for number of episodes
  while s is not a terminal state
    choose a from s using action selection policy
    observe R and s', update Q, update s
    update model M(s,a) = (R,s')
    repeat n times
      choose (s,a) to simulate
      simulate action: observe R and s'
      update Q
```

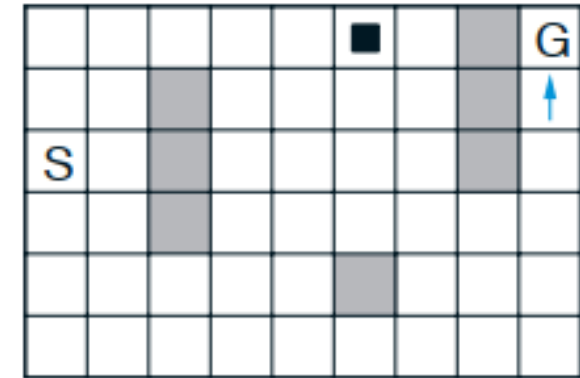


Model-based learning *Dyna-Q*

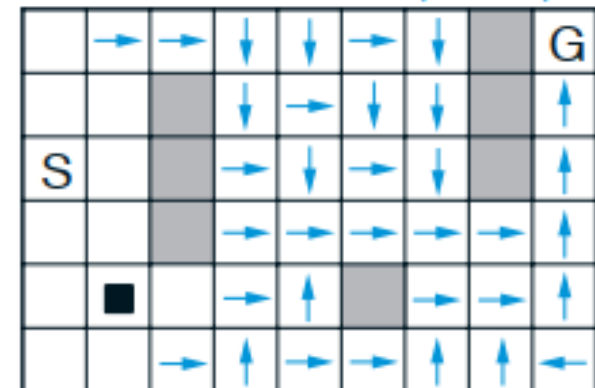
policy during episode 2



WITHOUT PLANNING ($n=0$)



WITH PLANNING ($n=50$)



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What can we do with a
reinforcement learning model?

What can we do with a reinforcement learning model?

Simulation

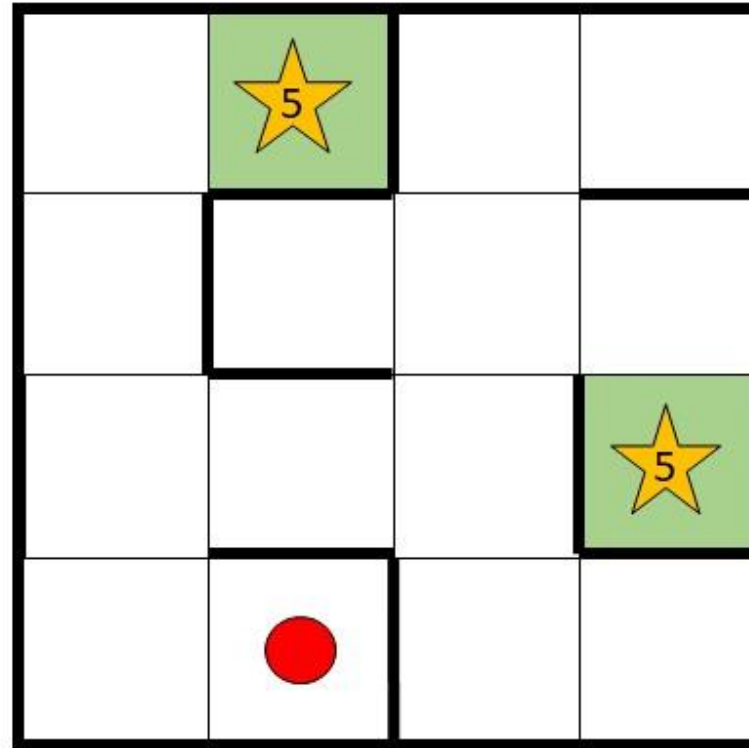
Model fitting

Model comparison

Simulation

- Set parameter values
- Let model run and produce results
- Can validate the model
 - how does the model behave with a high learning rate?

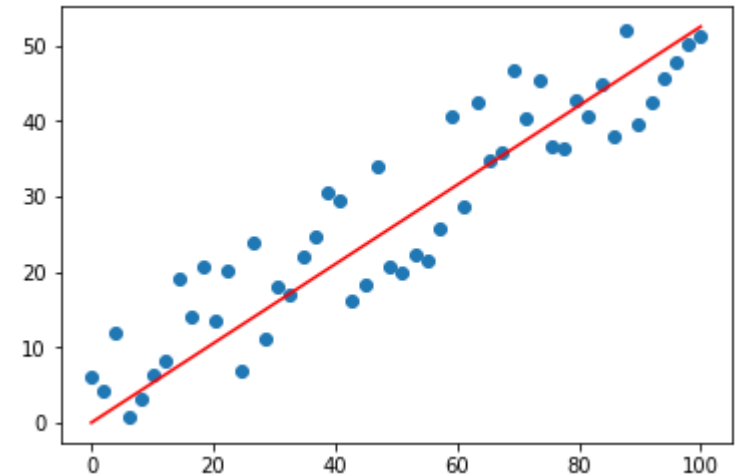
Simulation example – computer lab!



Model fitting

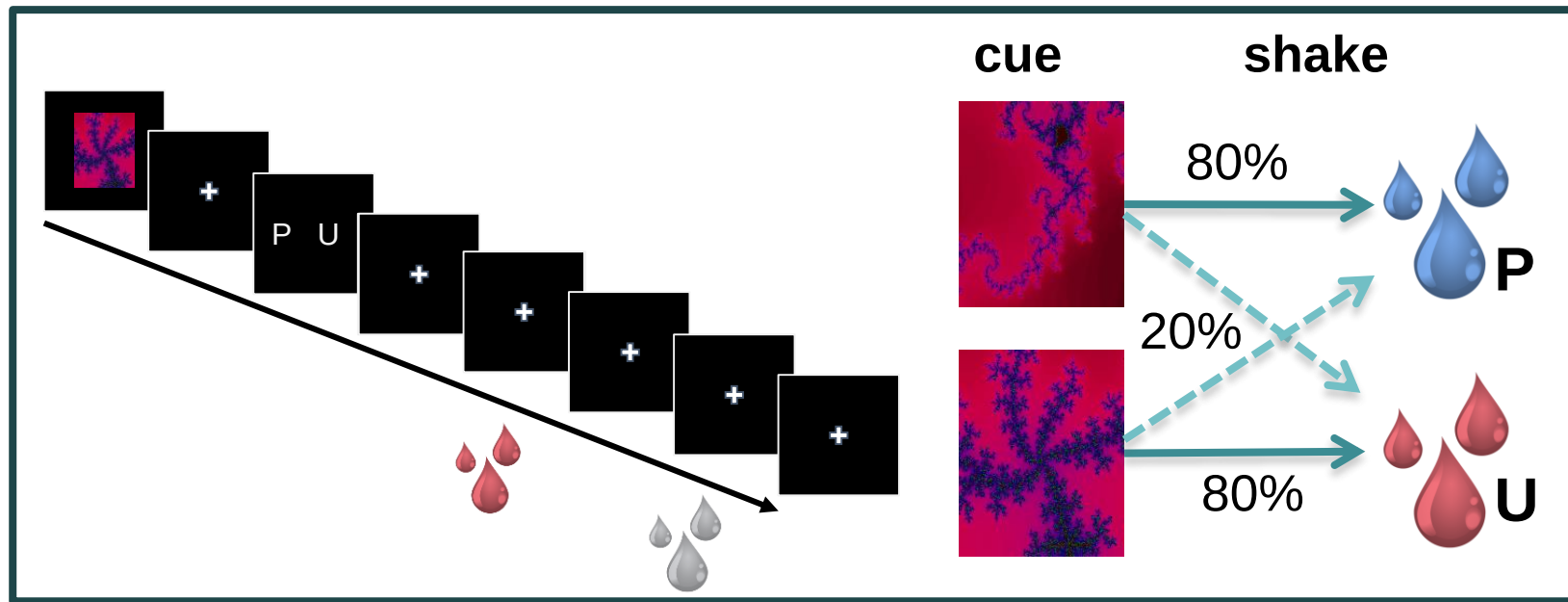
- Do an experiment
- Observe **behavior** (sequences of actions)
- Estimate the **parameters** of the model (e.g. α , γ , ε) such that the model behaves most similar to the observed data
- Recover **hidden states** of the model (e.g. Q-values and prediction errors) to learn about cognitive processes that are not directly observable

Simplest type of model fitting: Linear regression

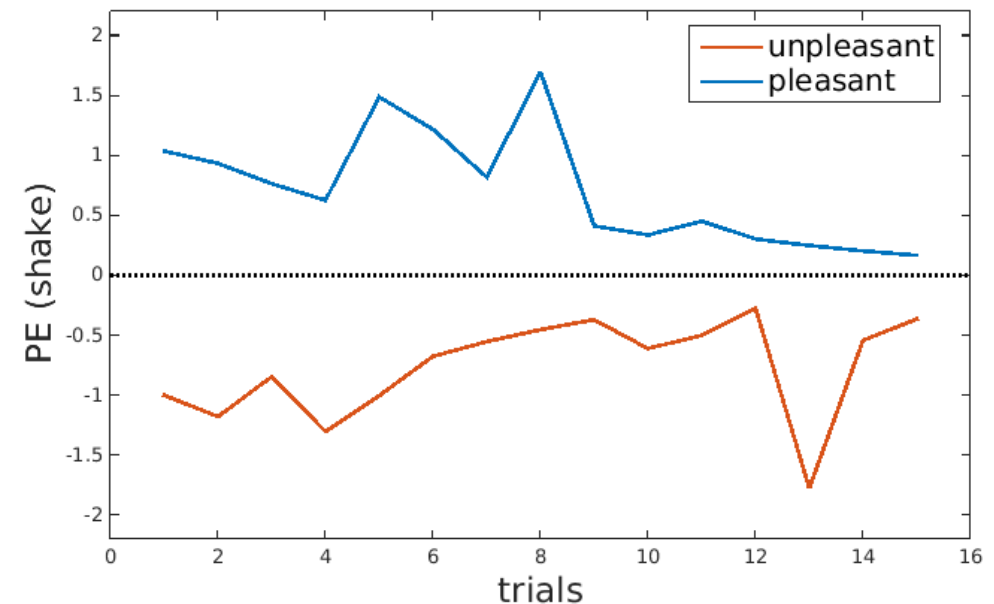
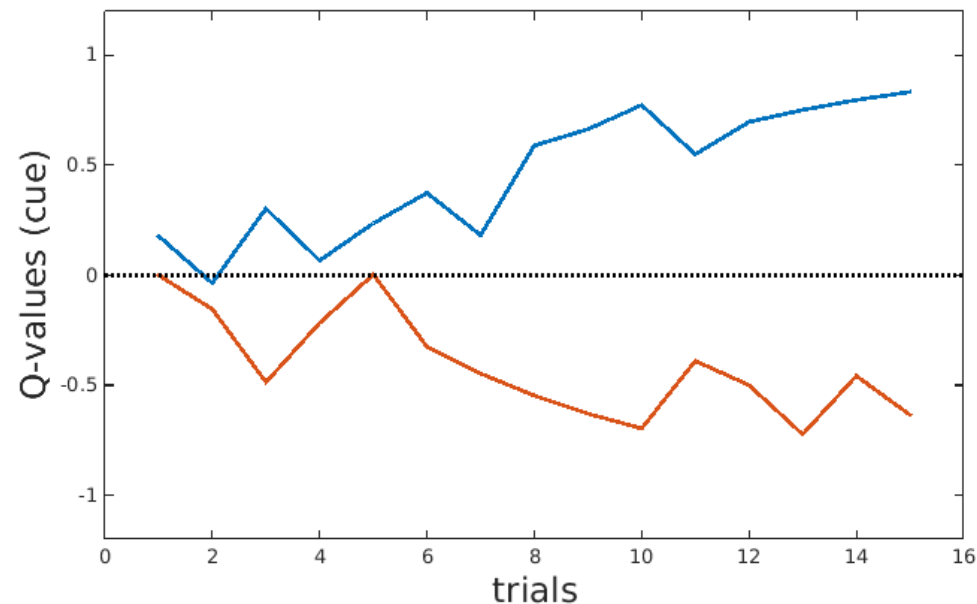


Model fitting example – milkshake experiment

RQ: How are positive and negative food prediction errors represented in the brain?

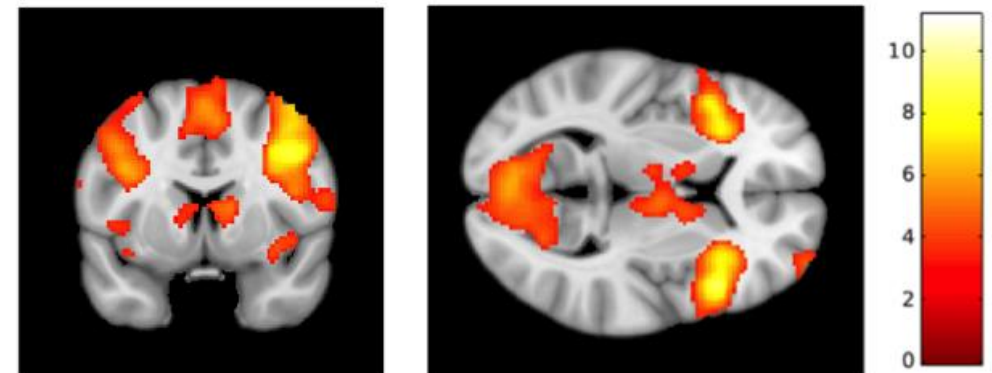


Model fitting example – milkshake experiment



Use prediction errors as parametric modulators on fmri data

→ Activity in caudate nucleus correlates with PE, not in nucleus accumbens



correlation with magnitude of prediction error

Excursion: Model fitting – how does it work?

I – inputs D – observed states/ our data ϵ – noise
 x – hidden states θ – model parameters m – our model

evolution function: $f(I, \theta) = x$ ***Q-value update***
observation function: $g(x, \theta) + \epsilon = D$ ***action selection function***

$p(\epsilon) \sim \mathcal{N}(\mathbf{0}, \sigma)$ noise is normally distributed

$p(D|m, \theta) = p(d_1, d_2 \dots |m, \theta) \sim \mathcal{N}(g(x, \theta), \sigma)$ model likelihood

Find the parameters θ that maximize the **model likelihood** by **gradient ascent**

Excursion: Model fitting – how does it work?

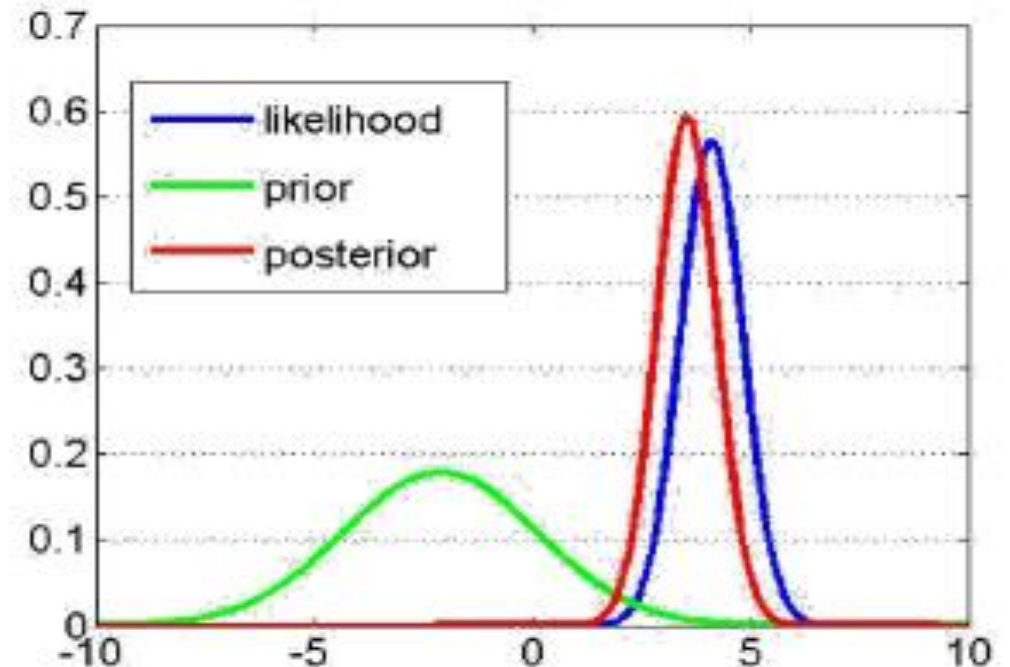
The Bayesian version

In Bayesian statistics, we are not interested in the model likelihood $p(D|\theta, M)$, but the posterior probability $p(\theta|D, M)$, which depend on the prior $p(\theta|M)$

$$p(\theta|D, M) = \frac{p(D|\theta, M) * p(\theta|M)}{p(D|M)}$$

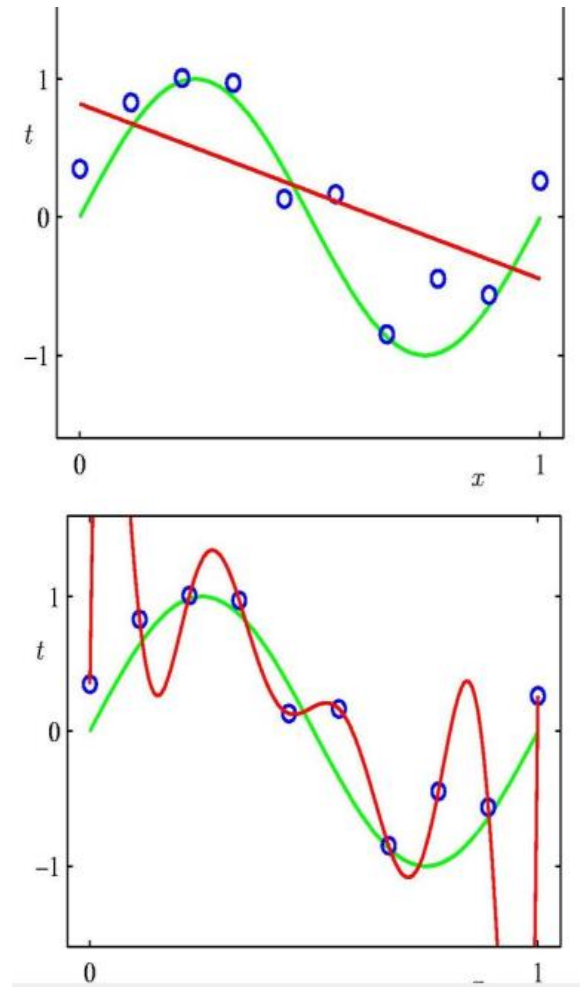
Find the parameters θ that maximize the **posterior**

[Click here for a nice tutorial & toolbox on this](#)



Model comparison

- Compare which of several models fits the data best (has the best model evidence)
- Penalize for number of parameters (avoid overfitting)
- Model comparison is a useful way to test hypotheses!
 - Formalize each hypothesis as a model
 - Test which model fits best



Model comparison

Example RQ: **Do humans show goal directed learning or instrumental conditioning in a maze experiment?**

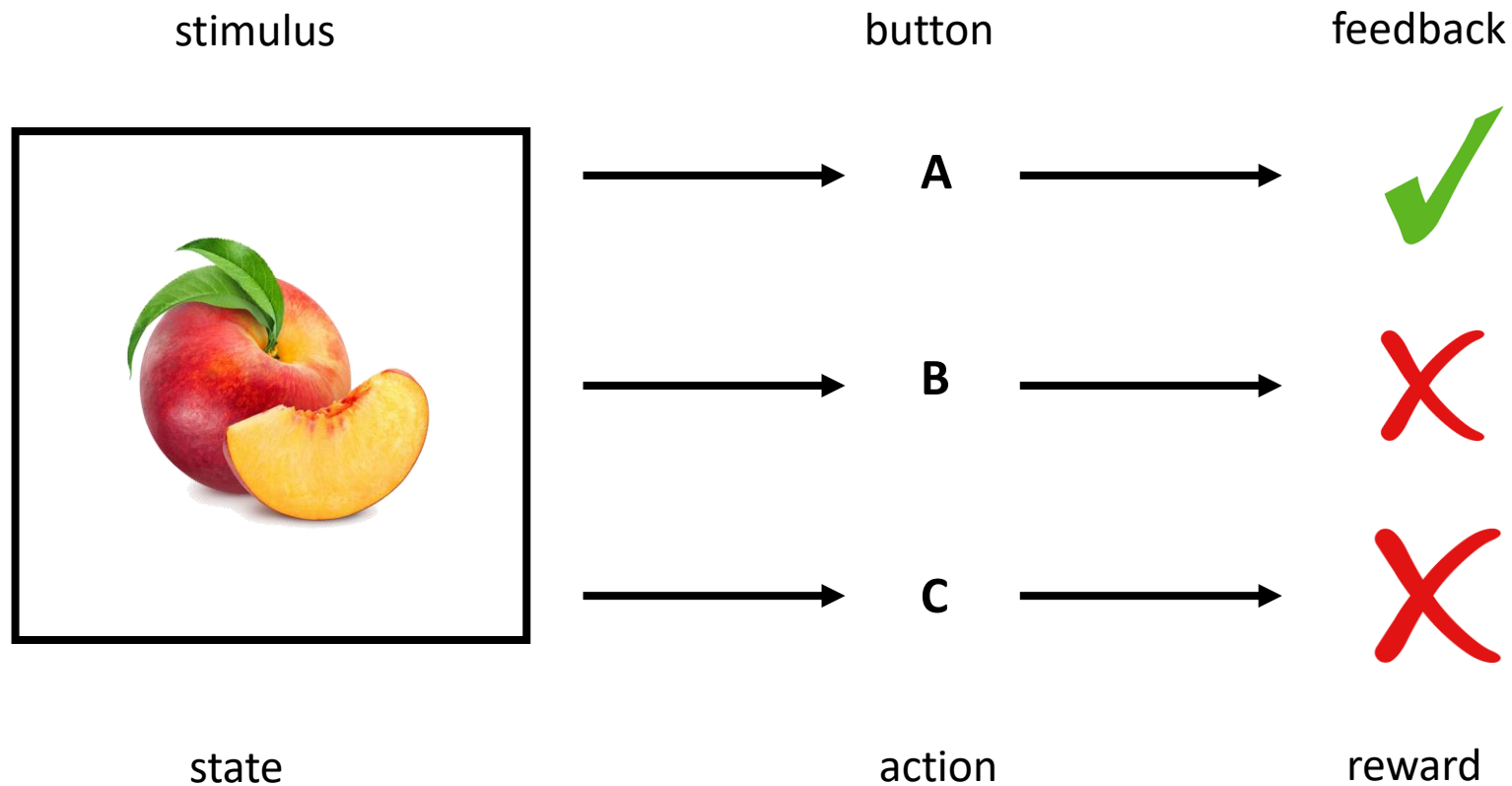
Goal directed learning → model based RL

Instrumental conditioning → Q-learning

Fit both models and see which one fits best!

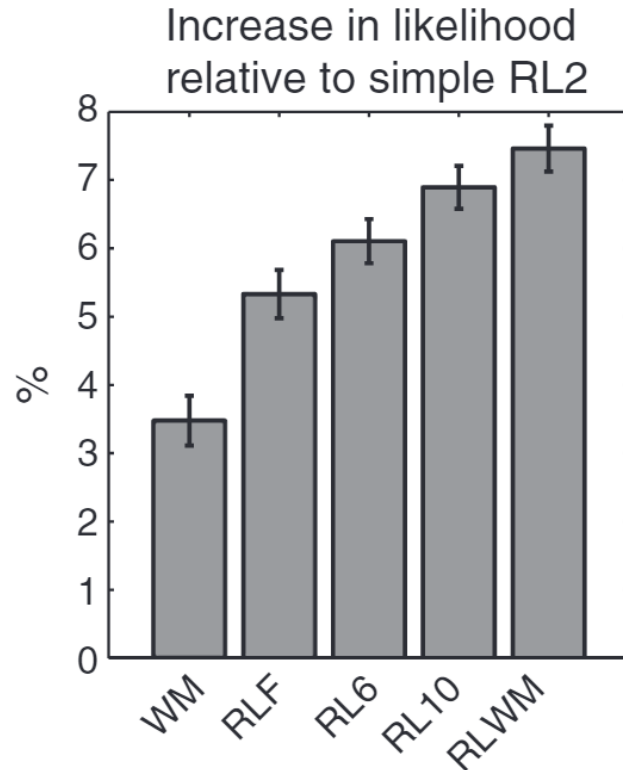
Model comparison example – n-armed bandit task

RQ: What is the role of working memory in reinforcement learning?



Model comparison example – n-armed bandit task

Collins & Frank 2012



RL2: simple RL model (2 parameters)

WM: working memory model (3 parameters)

RLF: RL model with forgetting parameter (3 parameters)

RL6: RL model with varying learning rate per set size (6 parameters)

RL10: RL model with varying learning rate and temperature (10 parameters)

RLWM: combined RL + working memory model (6 parameters)

→ WM component in RL, that is not accounted for in classical RL model

→ WM and RL components might account for prefrontal / basal ganglia activity respectively?

Excursion: Model comparison – how does it work?

To compare models, we need the **marginal likelihood** or **model evidence** $p(\mathbf{D}|\mathbf{m})$

$$p(\mathbf{D}|\mathbf{m}) = \int p(\mathbf{D}|\boldsymbol{\theta}, \mathbf{M}) * p(\boldsymbol{\theta}|\mathbf{M}) d\boldsymbol{\theta} \text{ marginalizing out the parameters}$$

Model evidence is a tradeoff between goodness of fit and model complexity

Can be used to compute Bayes factor:

$$\frac{p(\mathbf{D}|\mathbf{m}_1)}{p(\mathbf{D}|\mathbf{m}_2)} = \mathbf{BF}_{12} \quad \text{if } \mathbf{BF}_{12} > 20 \text{ – strong evidence for model 1}$$

[Click here for a nice tutorial & toolbox on this](#)

Extensions of RL algorithms

- Eligibility traces

update not only last state-action pairs, but all previously visited state-action pairs with decay factor λ

- Decaying learning rate

α decreases over time

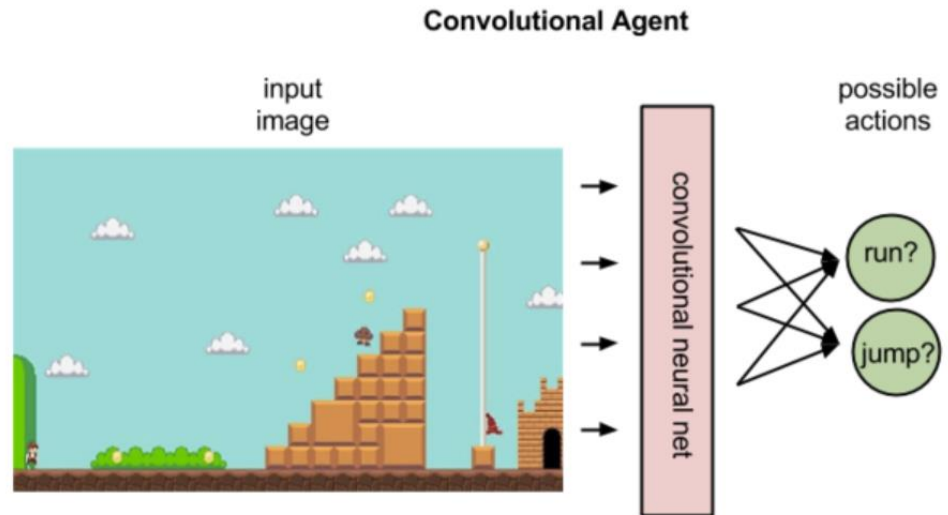
- Hybrid models

Model-based and model-free component

...

Deep reinforcement learning

- Deep reinforcement learning
 - Learn Q-values with deep neural nets



A nice tutorial on deep reinforcement learning